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Greater China Semiconductors | Asia Pacific

Old Memory: Divergent Trends in the Near Term

Recent memory cycle debate + CSP FCF concern + China memory position offload have caused a derating of old memory stocks, but we see stronger DDR4 pricing hikes and unprecedented legacy flash pricing power into 3Q. We think 1.0x 2028E BVPS offers good downside support.

Changing tide on mainstream memory: According to Trendforce ([Exhibit 1](#)), 3Q26 conventional DRAM mixed pricing is expected to increase by 13-18% Q/Q. Specifically, PC DDR5 pricing is seen rising by 15-20% Q/Q, server DDR5 pricing by 13-18% Q/Q, mobile LPDDR5(X) by 8-13% Q/Q, and NAND pricing by 10-15% Q/Q. Further, for 4Q26, Trendforce estimates price hikes of 3-8% Q/Q for conventional DRAM and 0-5% Q/Q for NAND. These estimated growth rates seem lower than some investors previously expected.

Yet, recently, we've seen stronger legacy memory pricing: Similar to our prior note ([link](#)), we believe DDR4 could enjoy continued monthly pricing hikes in 3Q. Compared to our pricing expectations one month ago, we believe CSPs are willing to offer higher pricing on DDR4 and may even try to sign a 1-2 year LTA (volume only) with suppliers. As a result, we believe monthly pricing could rise by 20%+ in 3Q, suggesting the Q/Q pricing hike could be 30% at least. We also continue to expect a 50-60% Q/Q price hike for SLC/MLC NAND in 3Q, and a 30-40% Q/Q price hike for NOR flash in 3Q.

Position offload in A share likely to see some stabilization: GigaDevice issued an announcement ([link](#)) that the company's niche memory products may see substantial price declines in the future. We see this warning as a potential longer term trend, rather than a near-term possibility, as we continue to see a structural supply gap across DDR4, SLC/MLC NAND, and NOR flash, supporting continued strong price hikes.

What is the downside support? GigaDevice (603986.SS): as a design house, it is hard to gauge the P/E floor, but the stock has already corrected 45.6% from its peak of Rmb84.0 as of Jun 29, providing some valuation support. Macronix/Nanya/Winbond trade at 1.5x/1.2x/1.2x 2028E BVPS, respectively; 1.0x historically has been a strong downside support level for greater China IDM companies.

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific

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Attractive

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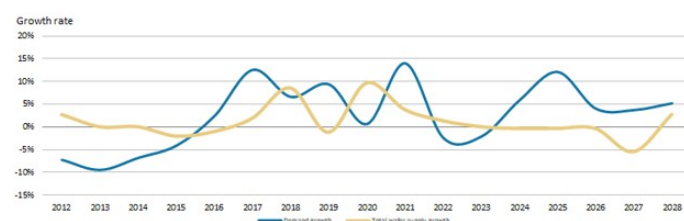
Key Charts and Exhibits

Exhibit 1: Latest pricing forecast for DRAM and NAND

	1Q26	2Q26	3Q26E - Old	3Q26E - New	4Q26E - Old	4Q26E - New
PC DRAM	DDR4: up 120-125% DDR5: up 110-115% Blended ASP: up 110-115%	DDR4: up 35-40% DDR5: up 45-50% Blended ASP: up 45-50%	up 3-8%	DDR4: up 15-20% DDR5: up 15-20% Blended ASP: up 15-20%	up 0-5%	up 3-8%
Server DRAM	DDR4: up 95-100% DDR5: up 93-98% Blended ASP: up 93-98%	DDR4: up 50-55% DDR5: up 53-58% Blended ASP: up 53-58%	up 10-15%	DDR4: up 13-18% DDR5: up 13-18% Blended ASP: up 13-18%	up 3-8%	up 3-8%
Mobile DRAM	LPDDR4X: up 58-63% LPDDR5(x): up 58-63%	LPDDR4X: up 73-78% LPDDR5(x): up 80-85%	up 3-8%	LPDDR4X: up 8-13% LPDDR5(x): up 8-13%	up 0-5%	up 0-5%
Graphics DRAM	GDDR6: up 95-100% GDDR7: up 95-100%	GDDR6: up 30-35% GDDR7: up 40-45%	GDDR6: up 3-8% GDDR7: up 10-15%	GDDR6: up 15-20% GDDR7: up 15-20%	GDDR6: up 0-5% GDDR7: up 3-8%	GDDR6: up 0-5% GDDR7: up 3-8%
Consumer DRAM	DDR3: up 75-80% DDR4: up 75-80%	DDR3: up 55-60% DDR4: up 55-60%	up 10-15%	DDR3: up 35-40% DDR4: up 28-33%	up 3-8%	up 3-8%
Total DRAM	Conventional DRAM: up 93-98% HBM Blended: up 83-88%	Conventional DRAM: up 58-63% HBM Blended: 53-58%	Conventional DRAM: up 8-13%	Conventional DRAM: up 13-18% HBM Blended: 8-13%	Conventional DRAM: up 3-8%	Conventional DRAM: up 3-8%
eMMC UFS	up 95-100%	up 75-80%	up 5-10%	up 5-10%	mostly flat	mostly flat
Enterprise SSD	up 75-80%	up 48-53%	up 13-18%	up 18-23%	up 3-8%	up 3-8%
Client SSD	up 105-110%	up 70-75%	up 5-10%	up 13-18%	mostly flat	mostly flat
3D NAND Wafers (TLC & QLC)	up 75-80%	up 28-33%	up 5-10%	up 5-10%	mostly flat	mostly flat
Total NAND Flash	up 85-90%	up 55-60%	up 8-13%	up 10-15%	up 0-5%	up 0-5%

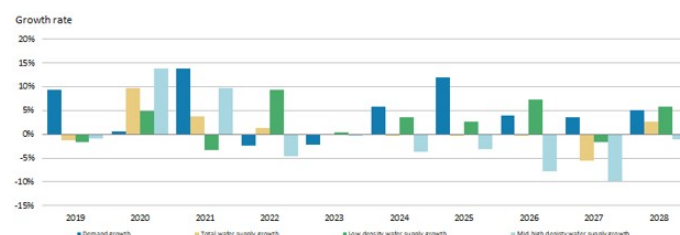
Source: Trendforce, Morgan Stanley Research estimates

Exhibit 2: NOR flash demand and supply growth rates

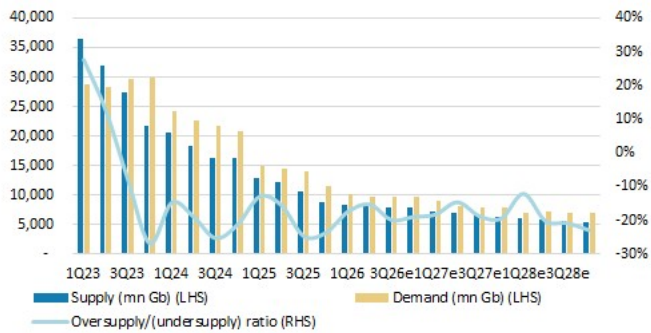


Source: Company data, Morgan Stanley Research

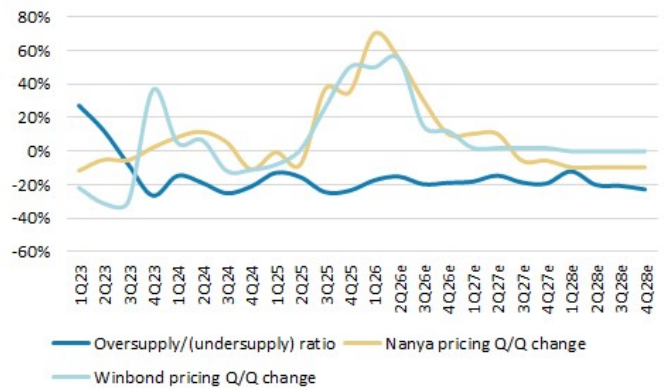
Exhibit 3: NOR flash demand growth and supply growth by density



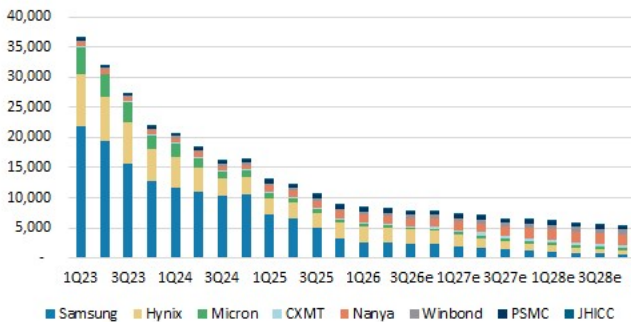
Source: Company data, Morgan Stanley Research

Exhibit 4: DDR4 quarterly supply and demand summary

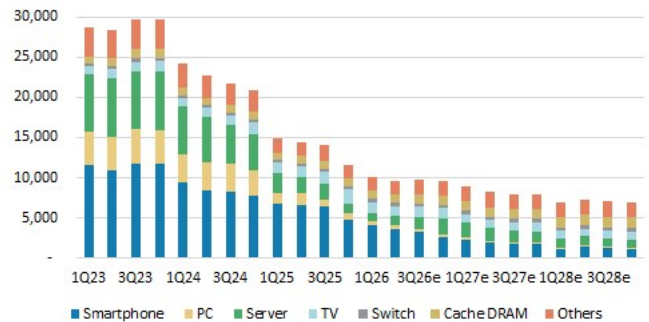
Source: Company data, IDC, Morgan Stanley Research (e) estimates

Exhibit 5: Quarterly oversupply/undersupply ratio vs. Nanya and Winbond pricing Q/Q change

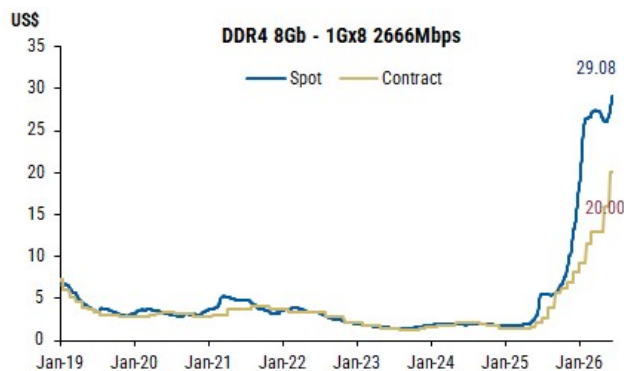
Source: Company data, IDC, Morgan Stanley Research (e) estimates

Exhibit 6: Quarterly supply breakdown (mn Gb)

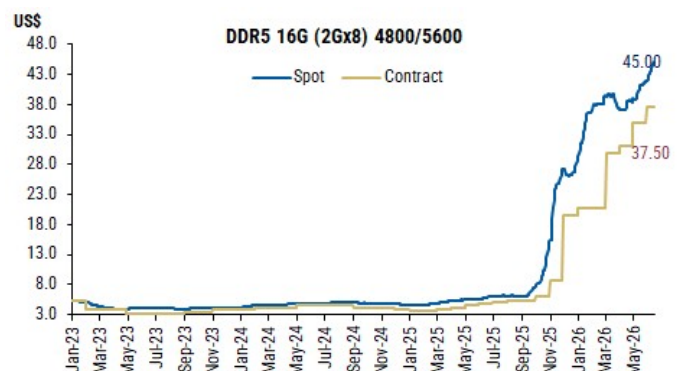
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 7: Quarterly demand breakdown by product (mn Gb)

Source: IDC, Morgan Stanley Research (e) estimates

Exhibit 8: DDR4 8Gb (1Gx8) pricing chart

Source: DRAMeXchange, Morgan Stanley Research. Data as of Jun 2026.

Exhibit 9: DDR5 16Gb (2Gx8) pricing chart

Source: DRAMeXchange, Morgan Stanley Research. Data as of Jun 2026.

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/27/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$84.32
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb399.77
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$121.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,460.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$608.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,241.02
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$4,050.00

GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,065.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$450.50
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$148.60
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$597.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$276.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb72.90
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,680.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb830.20
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$436.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb770.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb95.31
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$1,820.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb98.89
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$449.00
SMIC (0981.HK)	O (10/21/2025)	HK\$70.65
TSMC (2330.TW)	O (02/07/2022)	NT\$2,350.00
UMC (2303.TW)	O (05/19/2026)	NT\$126.00
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$157.00
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$355.00
Daisy Dai, CFA		
ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$161.60
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb60.74
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb92.22
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb32.33
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb313.59
Innoscence (2577.HK)	E (10/13/2025)	HK\$44.12
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb82.35
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$24.88
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb100.00
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb84.88
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb66.00
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb24.77
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb93.22
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$773.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,385.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$14,695.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$98.70
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$166.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb104.23
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb434.03
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$125.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$322.00
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb227.45
Novatek (3034.TW)	U (02/04/2026)	NT\$518.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$132.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$615.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$61.90
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$762.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb50.31
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$160.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$116.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$203.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb115.15

Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb373.90
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,035.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$598.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$12.62
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,110.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,720.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$270.90
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,395.00

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* Historical prices are not split adjusted.

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